

Angenommen, dass die Menge M aus n Elementen besteht und M auf M definiert ist. Dann gilt:

$$M \cup M = M, \quad M \cap M = M, \quad M \setminus M = \emptyset, \quad M \setminus \emptyset = M$$

Angenommen, dass die Menge M aus n Elementen besteht und M auf M definiert ist. Dann gilt:

$$\frac{1}{n} \sum_{i=1}^n x_i = \frac{1}{n} \sum_{i=1}^n y_i, \quad \frac{1}{n} \sum_{i=1}^n x_i = \frac{1}{n} \sum_{i=1}^n y_i$$

Angenommen, dass die Menge M aus n Elementen besteht und M auf M definiert ist. Dann gilt:

Angenommen, dass die Menge M aus n Elementen besteht und M auf M definiert ist. Dann gilt:

Angenommen, dass die Menge M aus n Elementen besteht und M auf M definiert ist. Dann gilt:



Es gilt: Das Ergebnis ist nicht positiv/negativ/nulldarstellend.

Angenommen, dass die Menge M aus n Elementen besteht und M auf M definiert ist.

$$\frac{1}{n} \sum_{i=1}^n x_i = \frac{1}{n} \sum_{i=1}^n y_i, \quad \frac{1}{n} \sum_{i=1}^n x_i = \frac{1}{n} \sum_{i=1}^n y_i \quad (100\%)$$

Angenommen, dass die Menge M aus n Elementen besteht und M auf M definiert ist.

$$\frac{1}{n} \sum_{i=1}^n x_i = \frac{1}{n} \sum_{i=1}^n y_i, \quad \frac{1}{n} \sum_{i=1}^n x_i = \frac{1}{n} \sum_{i=1}^n y_i \quad (100\%)$$

Also, notice that $\mu_{\mathcal{A}}(x) = 0$ if x is not in $\mathcal{A}$. So, $\mu_{\mathcal{A}}(x)$ is really just the indicator function for $\mathcal{A}$. We can write this as $\mu_{\mathcal{A}}(x) = \mathbb{1}_{\mathcal{A}}(x)$. So, $\mu_{\mathcal{A}}(x) = \mathbb{1}_{\mathcal{A}}(x)$ if $x \in \mathcal{A}$ and $\mu_{\mathcal{A}}(x) = 0$ if $x \notin \mathcal{A}$.

Definition 1.1 Let $\mathcal{A}$ and $\mathcal{B}$ be two sets. The *intersection* of $\mathcal{A}$ and $\mathcal{B}$ is the set $\mathcal{A} \cap \mathcal{B}$ consisting of all elements that are in both $\mathcal{A}$ and $\mathcal{B}$. The *union* of $\mathcal{A}$ and $\mathcal{B}$ is the set $\mathcal{A} \cup \mathcal{B}$ consisting of all elements that are in either $\mathcal{A}$ or $\mathcal{B}$.



Suppose that $\mathcal{A}$ and $\mathcal{B}$ are two sets. The *difference* of $\mathcal{A}$ and $\mathcal{B}$ is the set $\mathcal{A} \setminus \mathcal{B}$ consisting of all elements that are in $\mathcal{A}$ but not in $\mathcal{B}$.

The *symmetric difference* of $\mathcal{A}$ and $\mathcal{B}$ is the set $\mathcal{A} \oplus \mathcal{B}$ consisting of all elements that are in either $\mathcal{A}$ or $\mathcal{B}$ but not in both.



Let $\mathcal{A}$ and $\mathcal{B}$ be two sets. The *complement* of $\mathcal{A}$ is the set $\mathcal{A}^c$ consisting of all elements that are not in $\mathcal{A}$.

Let $\mathcal{A}$ and $\mathcal{B}$ be two sets. The *relative complement* of $\mathcal{A}$ with respect to $\mathcal{B}$ is the set $\mathcal{B} \setminus \mathcal{A}$ consisting of all elements that are in $\mathcal{B}$ but not in $\mathcal{A}$. The *relative complement* of $\mathcal{B}$ with respect to $\mathcal{A}$ is the set $\mathcal{A} \setminus \mathcal{B}$ consisting of all elements that are in $\mathcal{A}$ but not in $\mathcal{B}$. The *relative symmetric difference* of $\mathcal{A}$ and $\mathcal{B}$ is the set $\mathcal{A} \oplus \mathcal{B}$ consisting of all elements that are in either $\mathcal{A}$ or $\mathcal{B}$ but not in both. The *relative intersection* of $\mathcal{A}$ and $\mathcal{B}$ is the set $\mathcal{A} \cap \mathcal{B}$ consisting of all elements that are in both $\mathcal{A}$ and $\mathcal{B}$. The *relative union* of $\mathcal{A}$ and $\mathcal{B}$ is the set $\mathcal{A} \cup \mathcal{B}$ consisting of all elements that are in either $\mathcal{A}$ or $\mathcal{B}$.

$$\mathcal{A} \cap \mathcal{B} = \mathcal{A} \cap \mathcal{B} \quad \mathcal{A} \cup \mathcal{B} = \mathcal{A} \cup \mathcal{B}$$

Let $\mathcal{A}$ and $\mathcal{B}$ be two sets. The *relative difference* of $\mathcal{A}$ with respect to $\mathcal{B}$ is the set $\mathcal{B} \setminus \mathcal{A}$ consisting of all elements that are in $\mathcal{B}$ but not in $\mathcal{A}$.

The *relative symmetric difference* of $\mathcal{A}$ and $\mathcal{B}$ is the set $\mathcal{A} \oplus \mathcal{B}$ consisting of all elements that are in either $\mathcal{A}$ or $\mathcal{B}$ but not in both.

$$\mathcal{A} \oplus \mathcal{B} = (\mathcal{A} \setminus \mathcal{B}) \cup (\mathcal{B} \setminus \mathcal{A}) \quad \mathcal{A} \cap \mathcal{B} = \mathcal{A} \cap \mathcal{B}$$

Let $\mathcal{A}$ and $\mathcal{B}$ be two sets. The *relative complement* of $\mathcal{A}$ with respect to $\mathcal{B}$ is the set $\mathcal{B} \setminus \mathcal{A}$ consisting of all elements that are in $\mathcal{B}$ but not in $\mathcal{A}$. The *relative complement* of $\mathcal{B}$ with respect to $\mathcal{A}$ is the set $\mathcal{A} \setminus \mathcal{B}$ consisting of all elements that are in $\mathcal{A}$ but not in $\mathcal{B}$.

